「Chapter 1.

#1.1.1. Let $f = x^2 y$ and $g = y \cdot \sin z$ be functions on $\mathbb{R}^3$. Express the followings;

a). fg^2

sol). $fg^2 = x^2 y \cdot y \sin z = (xy)^2 \sin z$.

b) $\frac{\partial f}{\partial x} g + \frac{\partial g}{\partial y} f = 2xy \cdot y \sin z + \sin z \, x^2 y = (\sin z) \cdot xy (x + 2y)$

c) $\frac{\partial^2 (fg)}{\partial y \partial z} = \frac{\partial}{\partial y} (x^2 y^2 \cos z) = 2x^2 y \cos z$.

d) $\frac{\partial}{\partial y} \sin f = \cos(x^2 y) \cdot x^2$.

#1.2.1. Let $V = (-2, 1, -1)$ and $W = (0, 1, 3)$.

a). Given $P \in \mathbb{R}^3$, express the tangent vector $3V_p - 2W_p$ as a linear combination of $U_1(P), U_2(P), U_3(P)$.

Sol). observe $V_p = (-2, 1, -1)_p = -2 \cdot (1,0,0)_p + 1 \cdot (0,1,0)_p + (-1) \cdot (0,0,1)_p$

$W_p = (0, 1, 3)_p = 0 \cdot (1,0,0)_p + 1 \cdot (0,1,0)_p + 3 \cdot (0,0,1)_p$,

So $3V_p - 2W_p = -6 \cdot (1,0,0)_p + 1 \cdot (0,1,0)_p + (-7) \cdot (0,0,1)_p$.

#1.2.2. Let $V = xU_1 + yU_2$ and $W = 2x^2 U_2 - U_3$.

Compute the vector field $W - xV$, and find its value at the point $P = (-1, 0, 2)$.

Proof. $W - xV = 2x^2 U_2 - U_3 - (x^2 U_1 + xy U_2)$

$= -x^2 U_1 + (2x^2 - xy) U_2 - U_3$.

Now, $(W - xV)(-1, 0, 2) = (1, 2, -1)_p$.

Example 3.3, sec 2.3.

Consider the unit-speed helix

$$\beta(s) = \left(a\cdot\cos\frac{s}{c},\ a\cdot\sin\frac{s}{c},\ \frac{b}{c}\cdot s\right) \quad \text{where } a>0,\ c=\sqrt{a^2+b^2}.$$

Figure out the Frenet Frame.

$$T(s)=\beta'(s)=\left(-\frac{a}{c}\cdot\sin\frac{s}{c},\ \frac{a}{c}\cdot\cos\frac{s}{c},\ \frac{b}{c}\right)$$

$$T'(s)=\beta''(s)=\left(-\frac{a}{c^2}\cos\frac{s}{c},\ -\frac{a}{c^2}\sin\frac{s}{c},\ 0\right)=-\frac{a}{c^2}\cdot\left(\cos\frac{s}{c},\ \sin\frac{s}{c},\ 0\right)$$

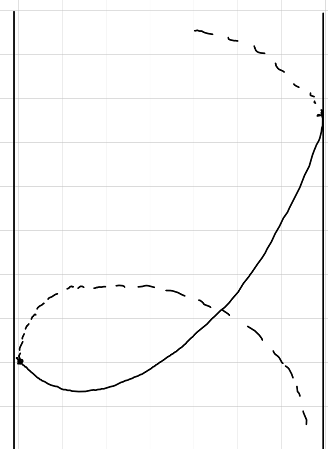
$$\kappa(s)=\|T'(s)\|=\frac{a}{c^2}=\frac{a}{a^2+b^2}>0.$$

$$N(s)=\frac{a^2+b^2}{a}\cdot T'(s)=\left(-\cos\frac{s}{c},\ -\sin\frac{s}{c},\ 0\right)$$

$$B(s)=\left(\frac{b}{c}\cdot\sin\frac{s}{c},\ -\frac{b}{c}\cdot\cos\frac{s}{c},\ \frac{a}{c}\right)$$

$$B'(s)=\left(\frac{b}{c^2}\cos\frac{s}{c},\ \frac{b}{c^2}\cdot\sin\frac{s}{c},\ 0\right)$$

$$\tau(s)=\frac{b}{c^2}=\frac{b}{a^2+b^2}.$$



Geometrically, if $b=0$, then the curvature $\kappa=\frac{1}{a}$, and the torsion is zero.

Sec 1.7 mapping.

#1~4, let $F(u,v) = (u^2-v^2, 2uv)$.

1). Find all points p s.t

a) $F(p) = (0,0)$

sol). $\begin{cases} u^2-v^2=0 \\ 2uv=0 \end{cases}$ $uv=0$ implies either $u=0$ or $v=0$, so $u=v=0$, by $u^2=v^2$.

c) $F(p)=p$

sol). $(u^2-v^2, 2uv) = (u,v)$

$2uv = v \Rightarrow v(2u-1)=0$. So either $v=0$ or $u=\frac{1}{2}$.

If $v=0$, $u^2=v^2=u$ so $u=0$ or 1

If $u=\frac{1}{2}$, then $v^2=\frac{1}{4}$, so $v=\pm\frac{1}{2}$,

So the tuples $(0,0)$, $(0,1)$, $(\frac{1}{2}, \pm\frac{1}{2})$ are the answer.

#3. Let $v=(v_1,v_2)$ and $p=(p_1,p_2)$. Express $F_*(v_p)$ directly.

sol). $F_*(v_p) = (v_p[u^2-v^2], v_p[2uv])$

for $f_1(u,v) = u^2-v^2$,

$$v_p[f_1] = \lim_{t\to0}\frac{f_1(p_1+tv_1, p_2+tv_2) - f(p_1,p_2)}{t}$$

$$= \lim_{t\to0}\frac{(p_1+tv_1)^2-(p_2+tv_2)^2-(p_1^2-p_2^2)}{t}$$

$$= \lim_{t\to0}\frac{2tp_1v_1-2tp_2v_2+t^2(v_1^2+v_2)^2}{t}$$

$$= 2p_1v_1 - 2p_2v_2$$

(or, by the lemma, $v_1\cdot 2u|_{u=p_1} + v_2\cdot(-2v|_{v=p_2}) = 2p_1v_1 - 2p_2v_2$.

And similarly $v_p[f_2] = v_1\cdot 2p_2 + v_2\cdot 2p_1$. Consequently,

$$F_*(v_p) = (2p_1v_1 - 2p_2v_2, 2p_2v_1 + 2p_1v_2)$$

#5. If $F: \mathbb{R}^n \to \mathbb{R}^m$ is linear, then $F_*(v_p) = F(v)_{F(p)}$

Proof. By the Theorem,

$$F_*(v_p) = (v_p[f_1], \cdots, v_p[f_m])_{F(p)}$$

And, since F is linear, each $f_i: \mathbb{R}^n \to \mathbb{R}$ also linear functional.

Therefore, $v_p[f_i] = \lim_{t \to 0} \dfrac{f_i(p+tv) - f_i(p)}{t} = \lim_{t \to 0} \dfrac{t \cdot f_i(v)}{t} = f_i(v).$ $(1 \le i \le m)$

so $F_*(v_p) = (f_1(v), \cdots, f_m(v))_{F(p)} = F(v)_{F(p)}.$

#7. If $v_p \in T_p\mathbb{R}^n$ and $g: \mathbb{R}^m \to \mathbb{R}$ is differentiable,
then for a mapping $F: \mathbb{R}^n \to \mathbb{R}^m$, $F_*(v_p)[g] = v_p[g(F)]$.
That is, a mapping preserves directional derivatives.

Proof. By the Theorem, for $F = (f_1, \cdots, f_m)$,

$$F_*(v_p) = (v_p[f_1], \cdots, v_p[f_m])_{F(p)},$$ so therefore

$$F_*(v_p)[g] = \sum_{i=1}^{m} v_p[f_i] \cdot \frac{\partial g}{\partial x_i}(F(p)).$$

And, by the chain rule,

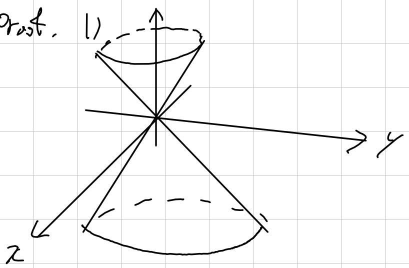
$$v_p[g(F)] = \sum_{i=1}^{m} v_i \frac{\partial g \circ F}{\partial x_i}(p) = \sum_{i=1}^{m} v_i \cdot \left[\frac{\partial g}{\partial F} \cdot \frac{\partial F}{\partial x_i}\right](p)$$

「Exercise 4.1.

#4.1.1. Prove that the followings are <u>Not</u> surfaces. (Formal proof is not required.)

1. Cone $z^2 = x^2+y^2$.
2. Closed disk; $x^2+y^2 \le 1$, $z=0$.
3. Folded plane: $xy=0$, $x \ge 0$, $y \ge 0$.

Proof. 1)



2. Describe a surface M; $ax+by+cz=d$ (a,b,c not all zero) by vector equation.

Sol).
$$M=\{(x,y,z)\in\mathbb{R}^3 \mid (a,b,c)\cdot(x,y,z)-d=0\}.$$
$$=\{(x,y,z)\in\mathbb{R}^3 \mid$$

#4.1.4.

Determine whether the following is a patch or not.

a). $X(u,v) = (u, uv, v)$

b). $X(u,v) = (u^2, u^3, v)$

c). $X(u,v) = (u, u^2, v+v^3)$

d). $X(u,v) = (\cos 2\pi u, \sin 2\pi u, v)$.

proof. a) is clearly 1-1, because $(u_1, u_1v_1, v_1) = (u_2, u_2v_2, v_2) \Rightarrow u_1 = u_2$ and $v_1 = v_2$.

To show the regularity, calculate the Jacobian:

$$DX = \begin{bmatrix} 1 & 0 \\ v & u \\ 0 & 1 \end{bmatrix}$$

clearly it has rank 2 for all $(u,v) \in \mathbb{R}^2$, so it is regular

b) is similarly 1-1, since $u_1^3 = u_2^3 \Rightarrow u_1 = u_2$. And, again,

$$DX = \begin{bmatrix} 2u & 0 \\ 3u^2 & 0 \\ 0 & 1 \end{bmatrix}$$

$v(v^2+1)$

it is has rank 1 when $u=0$, so it is Not regular.

c) is also 1-1, because $x \mapsto x + x^3$ is 1-1. And,

$$DX = \begin{bmatrix} 1 & 0 \\ 2u & 0 \\ 0 & 1+3v^2 \end{bmatrix}$$

is always rank 2, so regular.

d) is Not 1-1 on $\mathbb{R}^2$, but 1-1 on $(-1,1) \times \mathbb{R}$. Moreover,

$$DX = \begin{bmatrix} -2\pi \sin 2\pi u & 0 \\ 2\pi \cos 2\pi u & 0 \\ 0 & 1 \end{bmatrix}$$ is always rank 2

#4.1.5. a).

Prove that $M = \{(x,y,z) \in \mathbb{R}^3 \mid (x^2+y^2)^2 + 3z^2 = 1\}$ is a surface.

Proof. Observe that: for $(x,y,z) \in \mathbb{R}^3$,

$$(x^2+y^2)^2 = 1 - 3z^2 \text{ has solution only } -\frac{1}{\sqrt{3}} \leq z \leq \frac{1}{\sqrt{3}}$$

Construct four patches: the upper, the below, and the middle.

$$X_1 : D_1 \to \mathbb{R}^3 : (u,v) \mapsto \left(u, v, \frac{1}{\sqrt{3}} \cdot \sqrt{1-(u^2+v^2)^2}\right)$$

where $D_1 = \{(u,v) \mid u^2+v^2 < 1\}$.

$$X_2 : D_1 \to \mathbb{R}^3 : (u,v) \mapsto \left(u, v, -\frac{1}{\sqrt{3}}\sqrt{1-(u^2+v^2)^2}\right)$$

The X_1 and X_2 are Monge patch. And, to cover the middle circle,

$$X_3 : D_2 \to \mathbb{R}^3 : (u,v) \mapsto \left(\sqrt[4]{1-3v^2} \cdot \cos u, \sqrt[4]{1-3v^2} \cdot \sin u, v\right).$$

where $D_2 = (-\pi, \pi) \times \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

and $X_4 = X_3(u+\pi, v+\pi)$

Then, X_1, X_2, X_3, X_4 cover M.